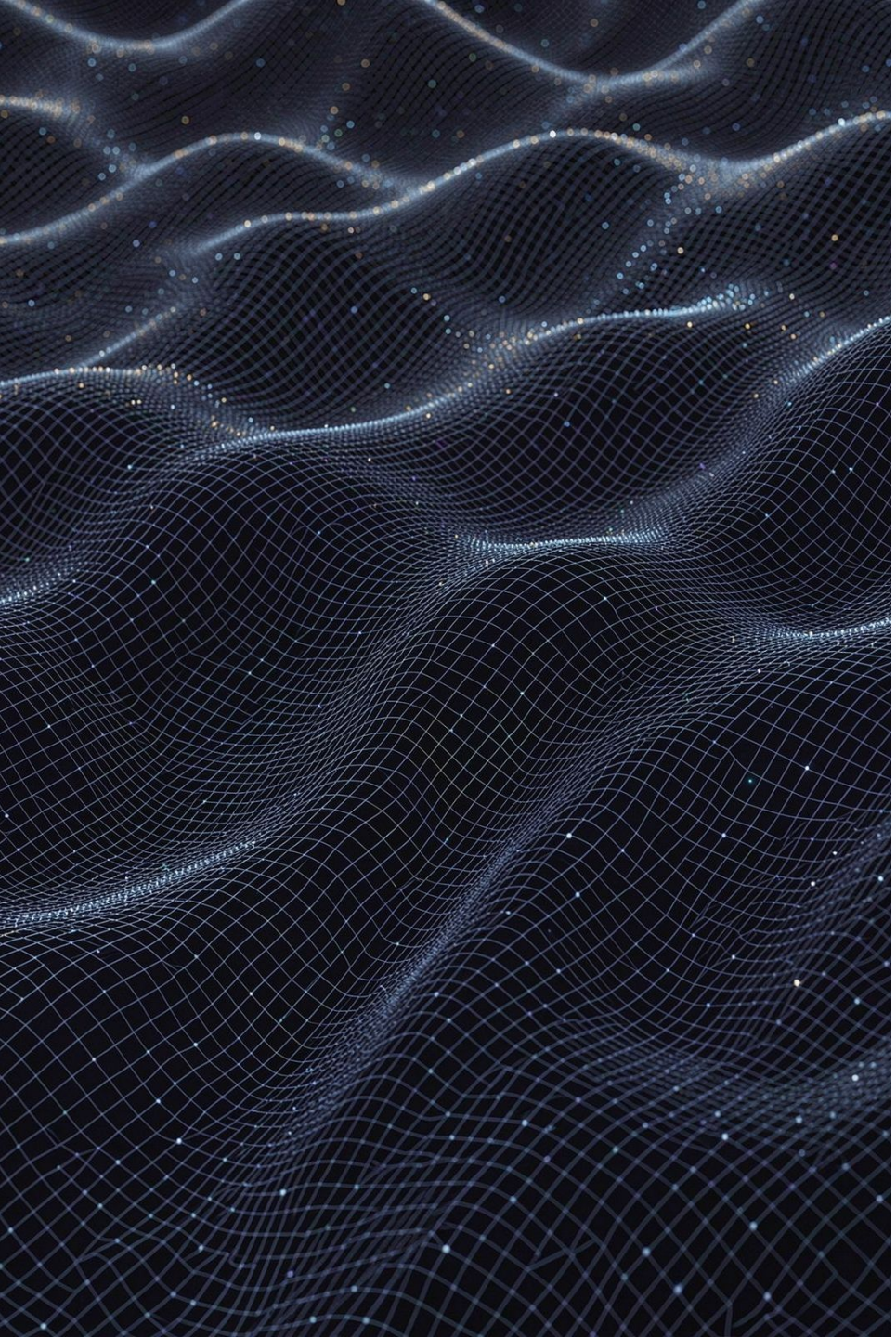


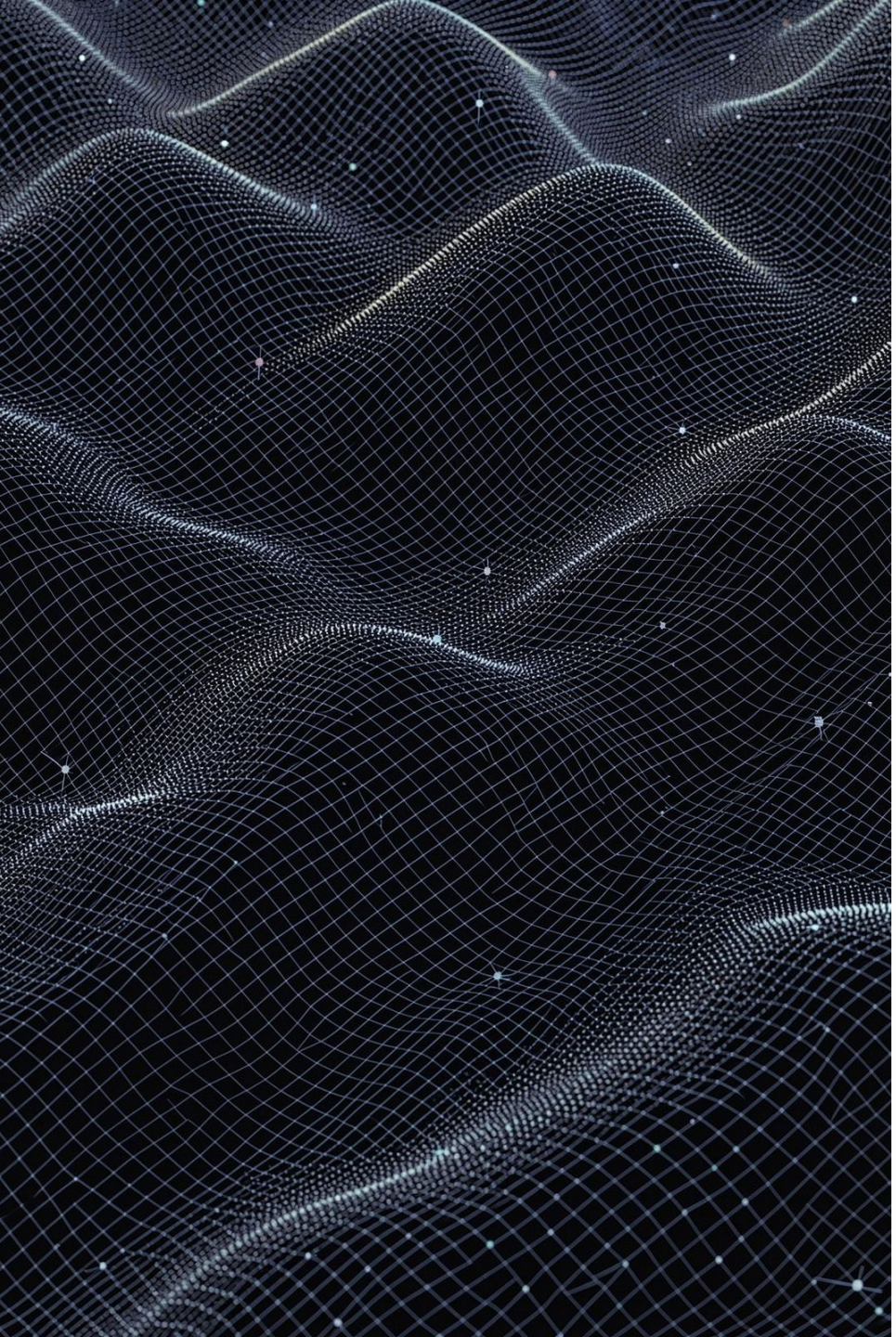
PROJECT 14: E-Commerce Order, Return & Refund Management System

Database Design and Implementation — Review 1 Presentation

Rayan Faheem Shaik - 25WU0102225

AI/ML Leopards

Made with GAMMA



Problem & Scope

The Problem

Online sellers lack integrated systems for products, inventory, carts, orders, payments, shipments, returns and refunds — causing overselling and delayed refund processing.

The Scope

Design a relational database covering:

- Customers & sellers
- Products & categories
- Inventory & carts
- Orders & order items
- Payments, shipments, returns, refunds & reviews

Made with GAMMA

Objectives



Unified database

Unified database for order management



Inventory control

Prevent overselling through inventory control



Returns and refunds

Enable efficient return and refund processing



Multiple roles

Support multiple user roles (customers, sellers, admins, support)



3NF normalization

Ensure data integrity with 3NF normalization

Made with GAMMA

Database Normalization: 1NF, 2NF, 3NF

1

1NF

Key rule: Atomic values only; no repeating groups.

Example: Store one phone number per row, not many in one field.

Builds the base: Removes lists and makes each cell hold a single value.

2

2NF

Key rule: Be in 1NF and remove partial dependencies.

Example: ProductName should depend on the full key, not just ProductID.

Builds on 1NF: Separates data that depends on only part of a composite key.

3

3NF

Key rule: Be in 2NF and remove transitive dependencies.

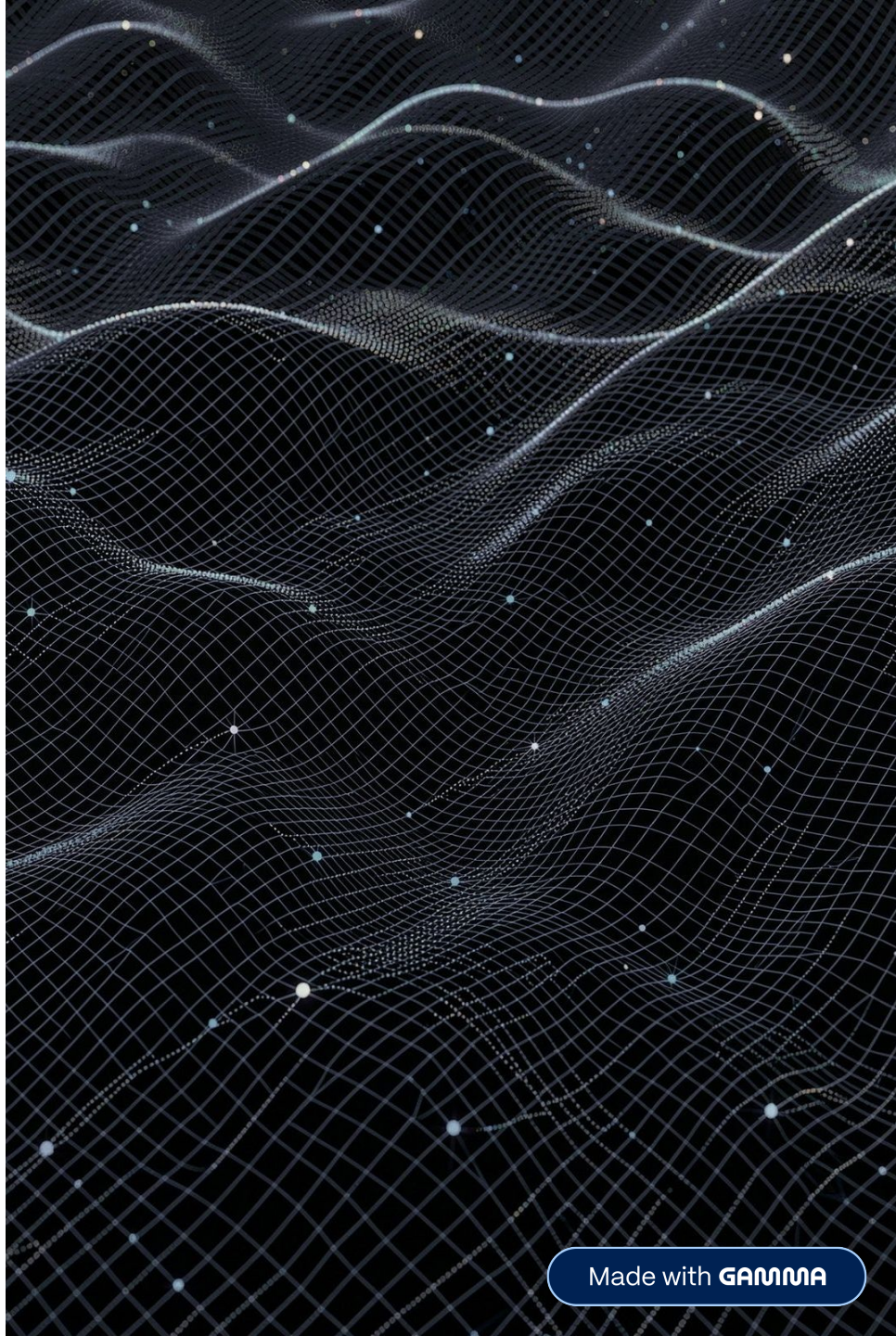
Example: Keep SellerCity in the Sellers table, not the Orders table.

Builds on 2NF: Ensures non-key fields depend only on the primary key.



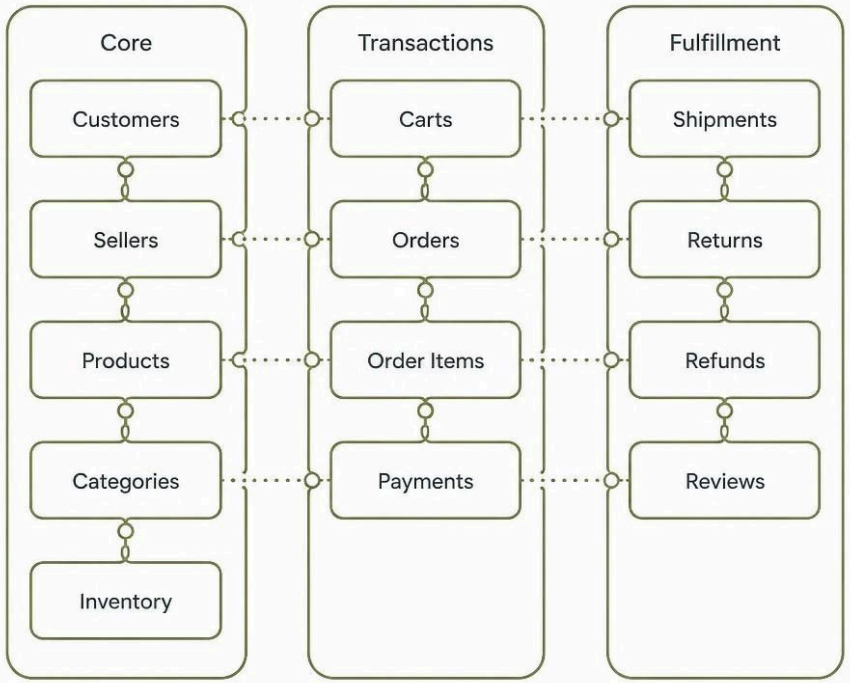
Our Project: 3NF Compliance

All tables follow 3NF normalization to reduce redundancy, prevent anomalies, and keep data consistent.



Made with GAMMA

Conceptual Design: Principal Entities (ER Diagram)



ER Diagram: 12 Principal Entities Across 3 Layers

This ER diagram covers all principal entities in the conceptual design: Customers, Sellers, Products, Categories, Inventory, Carts, Orders, Order Items, Payments, Shipments, Returns, Refunds, and Reviews.

Core Layer

- Customers, Sellers, Products, Categories, Inventory

Transaction Layer

- Carts, Orders, Order Items, Payments

Fulfillment Layer

- Shipments, Returns, Refunds, Reviews

Made with GAMMA

ER Diagram: Entity Relationships



Made with GAMMA

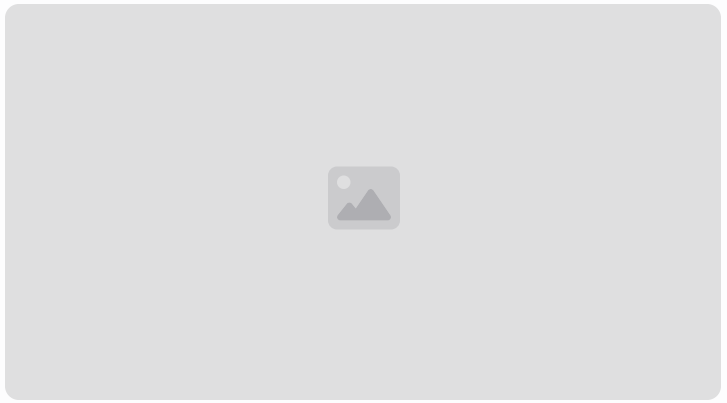
Initial Relational Schema



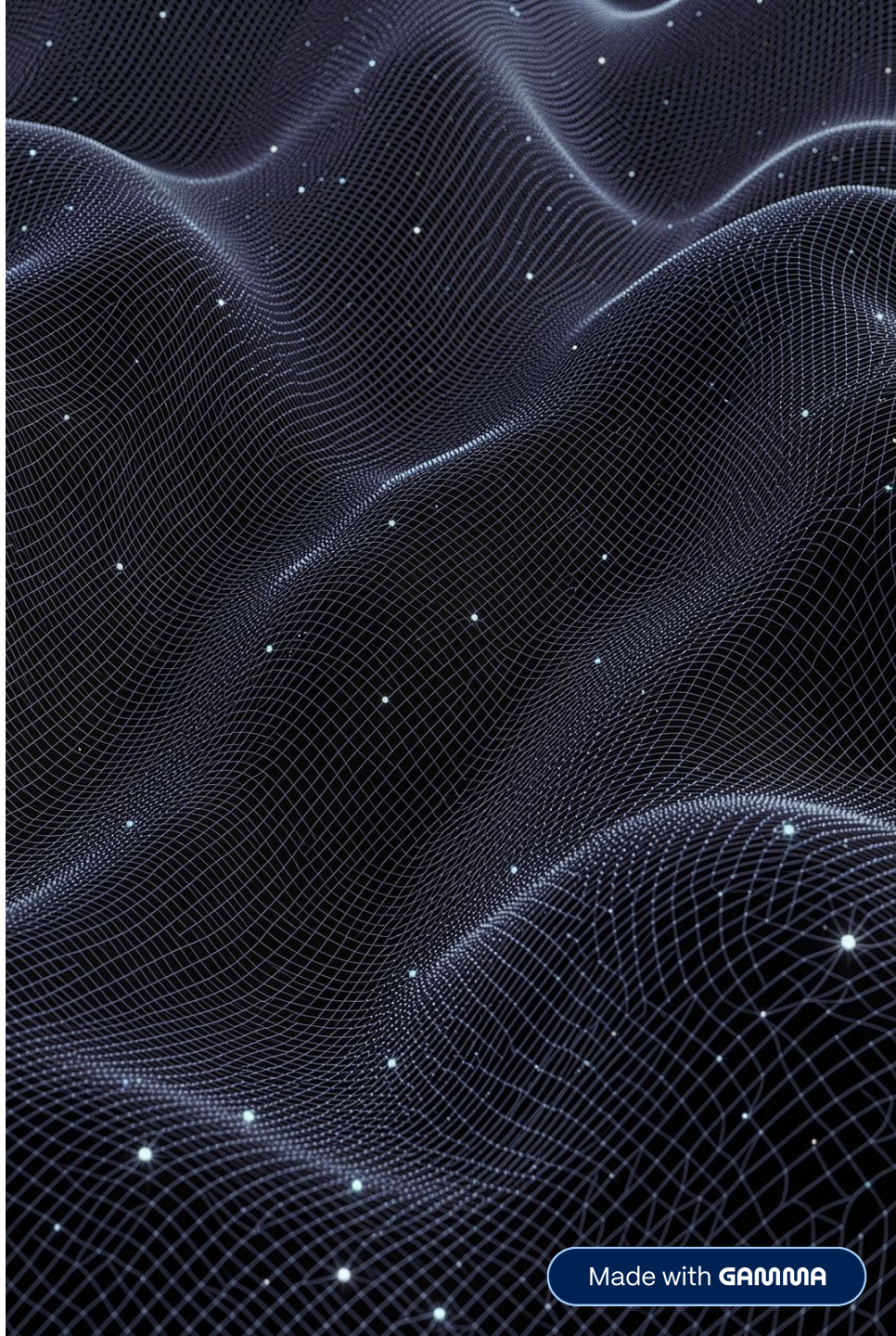
Table	Key Columns
CUSTOMERS	CustomerID *(PK)*, Email *(UNIQUE)*, Name, Phone, Address
PRODUCTS	ProductID *(PK)*, SKU *(UNIQUE)*, Price, SellerID *(FK)*, CategoryID *(FK)*

Made with GAMMA

Review 1 Deliverables Summary



- Problem identification and scope clearly defined
- Objectives aligned with business requirements
- Conceptual ER diagram with 13 principal entities
- Entity relationships documented
- Initial relational schema with primary and foreign keys
- Foundation ready for Review 2 implementation



Made with GAMMA